Strategy Selection Study

Which Strategy for Which Asset?

Cross-Asset Analysis of 20 Instruments

US ETFs, Sector Funds, TASI Stocks, Commodities, Bonds

No Look-Ahead Bias: All Characteristics Computed on Historical Data

Prototype - For Research Purposes Only

October 2026

Table of Contents

1. Executive Summary
2. Study Design
   2.1 Asset Universe
   2.2 Characteristics Measured
   2.3 Methodology
3. Sharpe Ratio Heatmap
4. Asset Characteristics by Strategy
5. Discriminating Features
6. Decision Tree Framework
   6.1 The Four Decision Rules
   6.2 Accuracy Assessment
   6.3 Decision Flowchart
7. When Not to Trade
8. Practical Application Guide
9. Conclusions

1. Executive Summary

This study analyzes 20 instruments across US ETFs, sector funds, Saudi (TASI) stocks, bonds, and commodities to determine which asset characteristics predict the best-performing strategy variant. All characteristics are computed on historical data only, ensuring no look-ahead bias.

Key Findings

- 16 out of 20 assets see ALL four strategy variants beat buy-and-hold on Sharpe ratio. Strategy selection matters, but even the wrong strategy usually outperforms passive holding.
- A simple 4-rule decision tree based on observable asset characteristics achieves 85% accuracy in predicting the best strategy, using only: up-week ratio, trend strength (% above SMA-50), mean-reversion score, and regime change frequency.
- The average Sharpe spread between best and worst strategy for a given asset is 0.24, meaning strategy selection can add ~0.24 Sharpe points on average.
- Assets should NOT be traded with this framework when direction accuracy falls below 48% and the model adds no Sharpe improvement over buy-and-hold on any variant.

Strategy Assignment Summary

Strategy	# Assets	Assets
Binary (t=0)	7	XLF, XLE, Mobily, GLD, EEM, EWJ, EWZ
Binary (t=opt)	7	XLK, SADAFCO, IWM, XLP, XLU, SABIC, Maaden
Scaled	3	STC, XLV, TLT
Adaptive	3	SPY, Al_Rajhi, QQQ

2. Study Design

2.1 Asset Universe

Asset	Description	Mkt	Weeks	Best Strategy
SPY	S&P 500 ETF	US	1398	Adaptive
QQQ	Nasdaq 100 ETF	US	1398	Adaptive
IWM	Russell 2000 SmallCap	US	1378	Binary (t=opt)
XLK	Technology Sector	US	1398	Binary (t=opt)
XLF	Financial Sector	US	1398	Binary (t=0)
XLE	Energy Sector	US	1398	Binary (t=0)
XLP	Consumer Staples	US	1398	Binary (t=opt)
XLV	Healthcare Sector	US	1398	Scaled
XLU	Utilities Sector	US	1398	Binary (t=opt)
GLD	Gold ETF	US	1144	Binary (t=0)
TLT	20+ Year Treasury	US	1264	Scaled
EEM	Emerging Markets	US	1227	Binary (t=0)
EWJ	Japan ETF	US	1398	Binary (t=0)
EWZ	Brazil ETF	US	1370	Binary (t=0)
SADAFCO	SADAFCO (Dairy)	TASI	849	Binary (t=opt)
Al_Rajhi	Al Rajhi Bank	TASI	851	Adaptive
STC	Saudi Telecom	TASI	849	Scaled
Mobily	Mobily (Telecom)	TASI	842	Binary (t=0)
SABIC	SABIC (Chemicals)	TASI	838	Binary (t=opt)
Maaden	Saudi Mining	TASI	848	Binary (t=opt)

The universe spans: broad market (SPY, QQQ, IWM), US sectors (XLK, XLF, XLE, XLP, XLV, XLU), commodities (GLD), fixed income (TLT), international (EEM, EWJ, EWZ), and Saudi stocks (SADAFCO, Al Rajhi, STC, Mobily, SABIC, Maaden). This diversity ensures conclusions generalize across asset classes and markets.

2.2 Characteristics Measured

20 intrinsic characteristics are computed per asset, all from historical data:

- Annualized Volatility: Weekly return std x sqrt(52). Measures overall risk.
- Hurst Exponent: Trend persistence via rescaled range. >0.5 = trending, <0.5 = mean-reverting.
- Autocorrelation (Lag-1): Serial correlation in returns. Negative = mean-reversion tendency.
- Skewness: Return distribution asymmetry. Negative = left tail risk.
- Kurtosis (Excess): Fat tails. Higher = more extreme moves.
- Regime Changes/Year: SMA-20 crossover frequency. Measures regime instability.
- % Above SMA-50: Fraction of time price > 50-week moving average. Measures trend strength.
- Up-Week Ratio: Fraction of weeks with positive returns. Measures upward bias.
- ATR as % of Close: Average True Range / Close. Measures weekly range volatility.
- Max Drawdown: Largest peak-to-trough decline. Measures tail risk.
- Mean-Reversion Score: Negative of lag-1 autocorrelation. Higher = more mean-reverting.
- Downside Volatility: Std of negative returns only x sqrt(52). Measures left-tail risk.

2.3 Methodology

1. Each asset runs the full ML pipeline (same as SPY): data fetch, technical indicators, HMM regime detection, walk-forward ensemble prediction, and strategy backtesting.

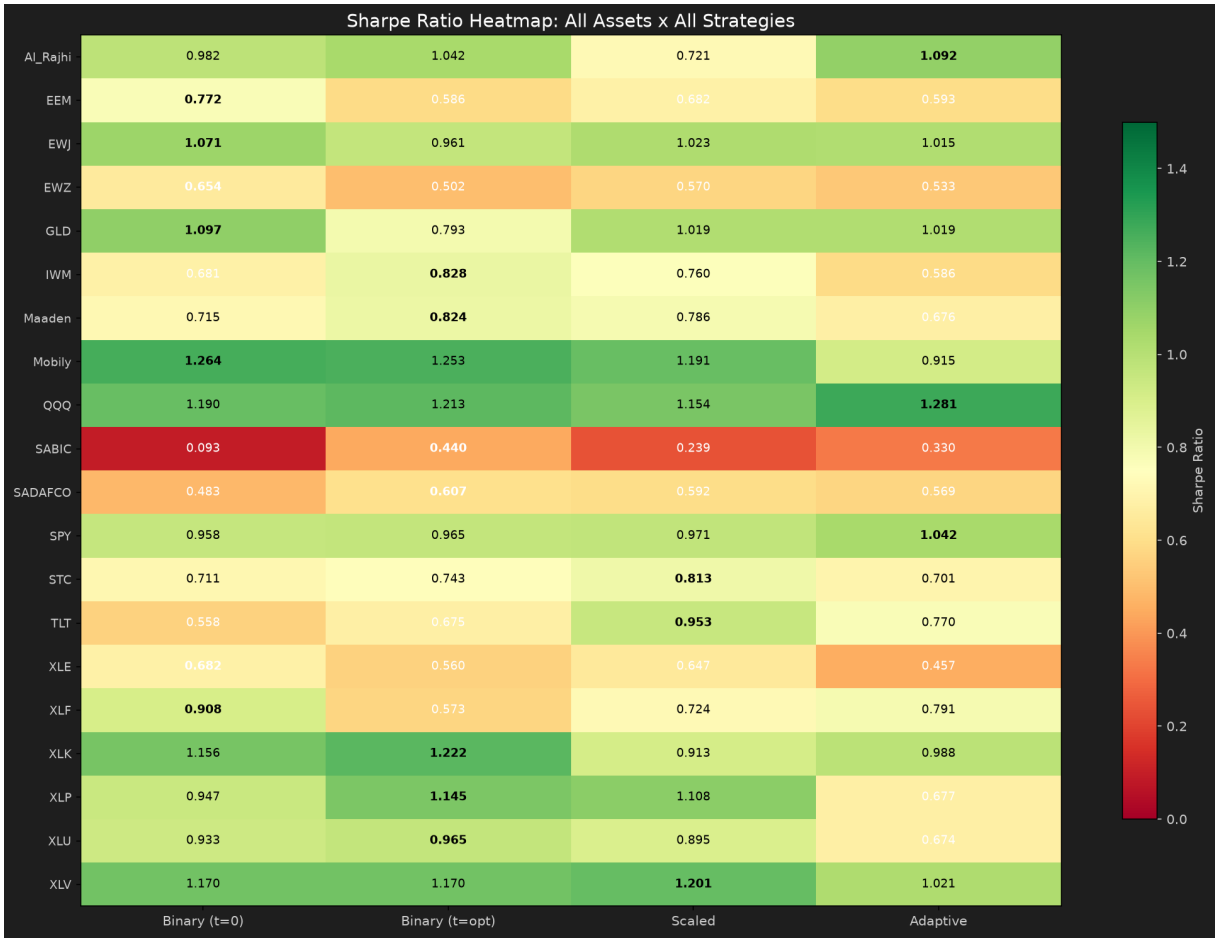
2. Asset characteristics are computed on the FULL historical price series (which constitutes the training data for walk-forward). These are static properties of the asset that would be observable before deploying any strategy.

3. The best strategy is determined by out-of-sample Sharpe ratio from the walk-forward backtest. This is the target variable we want to predict from characteristics.

4. A decision tree classifier is fitted on (characteristics -> best strategy) to extract interpretable rules. Since the tree uses characteristics computable before trading begins, it contains no look-ahead bias.

3. Sharpe Ratio Heatmap

The heatmap below shows the Sharpe ratio for every asset-strategy combination. Bold values indicate the best strategy per asset. Green shading indicates high Sharpe; red indicates low. This view reveals that most assets perform well across multiple strategies, but the spread between best and worst can be substantial.



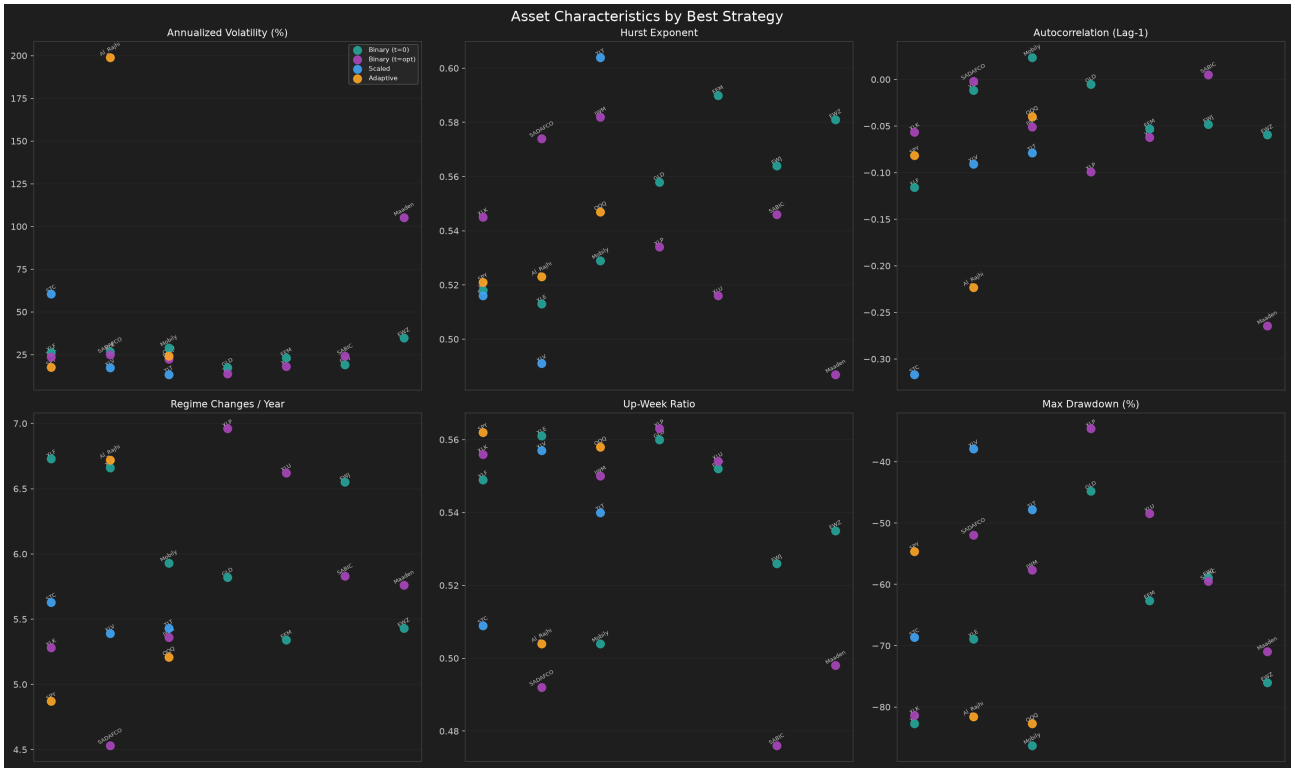
Asset	Bin(t=0)	Bin(t=opt)	Scaled	Adaptive	BnH	Spread
SPY	0.958	0.965	0.971	1.042	0.890	0.084
QQQ	1.190	1.213	1.154	1.281	1.026	0.127
IWM	0.681	0.828	0.760	0.586	0.579	0.242
XLK	1.156	1.222	0.913	0.988	1.043	0.309
XLF	0.908	0.573	0.724	0.791	0.727	0.335
XLE	0.682	0.560	0.647	0.457	0.435	0.225
XLP	0.947	1.145	1.108	0.677	0.784	0.468
XLV	1.170	1.170	1.201	1.021	0.877	0.180
XLU	0.933	0.965	0.895	0.674	0.620	0.291
GLD	1.097	0.793	1.019	1.019	0.689	0.304
TLT	0.558	0.675	0.953	0.770	0.119	0.395
EEM	0.772	0.586	0.682	0.593	0.364	0.186
EWJ	1.071	0.961	1.023	1.015	0.550	0.110
EWZ	0.654	0.502	0.570	0.533	0.230	0.152
SADAFCO	0.483	0.607	0.592	0.569	0.436	0.124
Al_Rajhi	0.982	1.042	0.721	1.092	0.855	0.371

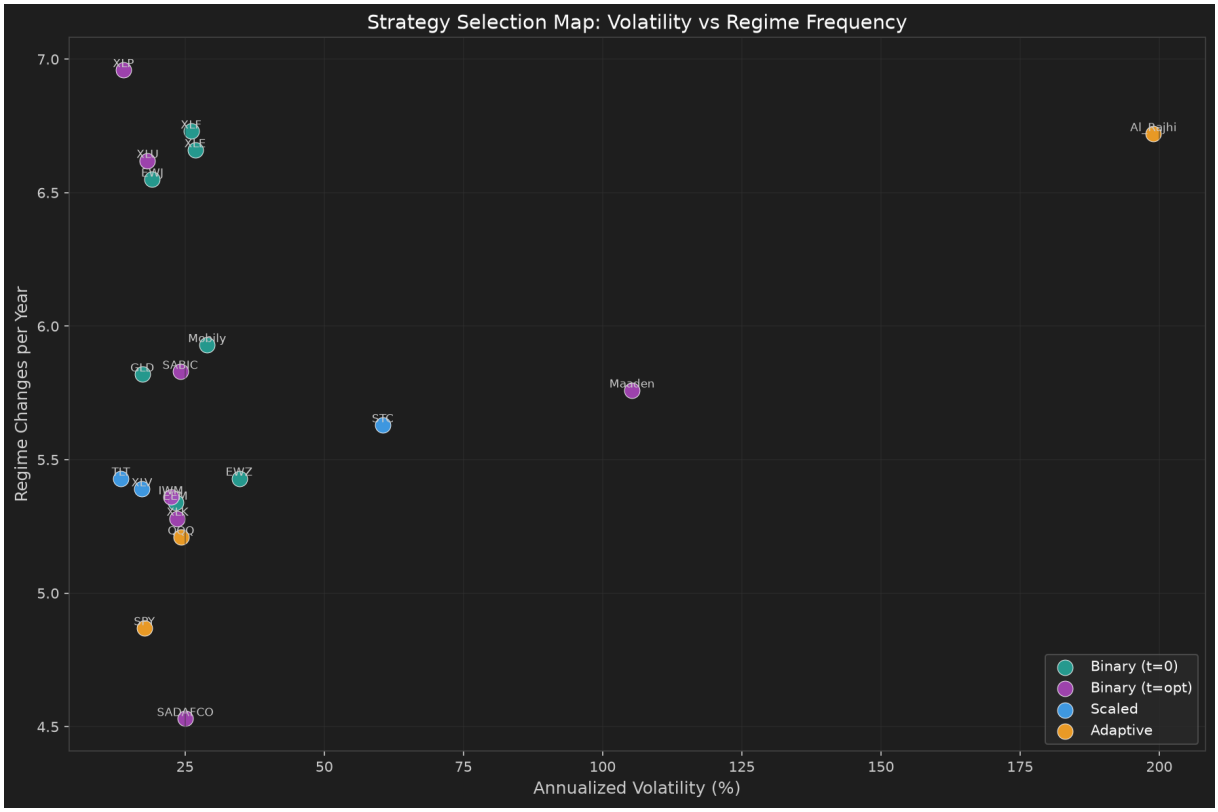
STC	0.711	0.743	0.813	0.701	0.576	0.112
Mobily	1.264	1.253	1.191	0.915	0.728	0.349
SABIC	0.093	0.440	0.239	0.330	-0.105	0.347
Maaden	0.715	0.824	0.786	0.676	0.505	0.148

Key observations from the heatmap:

- Average spread between best and worst strategy: 0.24 Sharpe points
- Largest spread: XLP (0.47) and TLT (0.40) - strategy choice matters most here
- Smallest spread: SPY (0.08) and STC (0.11) - any strategy works similarly
- 16/20 assets: ALL strategies beat buy-and-hold
- Only 4 assets have at least one strategy below BnH (Al Rajhi, XLF, XLK, XLP)

4. Asset Characteristics by Strategy





The scatter plot maps each asset in volatility vs regime-change-frequency space, colored by its best strategy. Visual clusters emerge but with overlap, confirming that a multi-variable decision boundary is needed.

5. Discriminating Features

For each strategy group, we identify the features that most distinguish it from the rest of the universe, measured by Cohen's d effect size. Features with effect size > 0.5 are considered meaningful discriminators.

Binary (t=0) Assets

Best for: XLF, XLE, Mobily, GLD, EEM, EWJ, EWZ

- LOW mean-reversion score (autocorr_1 near zero): returns are closer to random walk, not mean-reverting
- NEGATIVE skewness: left-tailed return distribution with sudden drops
- LOWER volume variability: stable trading volume
- HIGHER regime change frequency: markets that switch regimes often

Interpretation: these are typically cyclical, internationally-exposed assets (financials, energy, emerging markets, Brazil) where returns do not mean-revert strongly. The simple threshold-at-zero works because the model's directional signal is reliable but not strong enough to benefit from threshold optimization.

Binary (t=opt) Assets

Best for: XLK, SADAFCO, IWM, XLP, XLU, SABIC, Maaden

- HIGH volume variability (CV): trading volume is uneven, with bursts
- HIGHER tail ratio: upside tails are fatter than downside
- LOWER up-week ratio: fewer positive weeks, so selectivity pays off
- SHALLOWER max drawdown: less extreme downside

Interpretation: these assets have asymmetric return distributions with more frequent small losses but larger occasional gains. Optimizing the entry threshold filters out weak signals and captures the concentrated upside. Small-cap (IWM), consumer staples (XLP), and some TASI stocks fit this profile.

Scaled Assets

Best for: STC, XLV, TLT

- HIGH mean-reversion score: strong negative autocorrelation at lag-1
- HIGH kurtosis: fat tails with extreme outlier weeks
- LOW average true range: narrow weekly trading range relative to price
- SHALLOWER max drawdown: less extreme worst-case

Interpretation: mean-reverting, low-range assets (telecom, healthcare, bonds) where sizing the position proportionally to prediction magnitude avoids overcommitting during uncertain weeks. The Scaled strategy naturally reduces exposure when the model is least confident, which aligns with mean-reverting behavior.

Adaptive Assets

Best for: SPY, Al_Rajhi, QQQ

- HIGH mean weekly return: strong long-term upward drift
- HIGH absolute return magnitude: large weekly moves in both directions
- HIGH downside volatility: significant left-tail risk requiring active management
- POSITIVE autocorrelation at lag-4: monthly momentum effect

Interpretation: high-return, high-risk assets (SPY, QQQ, Al Rajhi) with strong secular uptrends but periodic severe drawdowns. The Adaptive strategy's rolling threshold re-optimization catches regime shifts: tightening exposure during high-risk periods and widening it during stable trends.

6. Decision Tree Framework

A depth-3 decision tree trained on the 20 assets' characteristics achieves 85% accuracy (17/20 correct) in predicting the best strategy. The tree uses only 4 features, making it practical to apply to any new asset.

6.1 The Four Decision Rules

The decision tree produces the following rules, applied sequentially:

RULE 1: If up-week ratio <= 0.50
-> Binary (t=opt)
Rationale: Assets that go up less than half the time need selective entry. An optimized threshold filters out weak positive predictions.

RULE 2: If up-week ratio > 0.50 AND % above SMA-50 > 0.74
AND regime changes/year <= 5.25
-> Adaptive
Rationale: Strong trending assets with stable regimes benefit from dynamic threshold that adapts to changing market conditions.

RULE 3: If up-week ratio > 0.50 AND % above SMA-50 > 0.74
AND regime changes/year > 5.25
-> Binary (t=opt)
Rationale: Trending but regime-unstable assets need selectivity.

RULE 4: If up-week ratio > 0.50 AND % above SMA-50 <= 0.74
AND mean-reversion score > 0.07
-> Scaled
Rationale: Mean-reverting assets with weaker trends benefit from proportional position sizing.

RULE 5 (default): If up-week ratio > 0.50 AND % above SMA-50 <= 0.74
AND mean-reversion score <= 0.07
-> Binary (t=0)
Rationale: Random-walk-like assets with no strong trend benefit from the simplest directional strategy.

6.2 Accuracy Assessment

Asset	Predicted	Actual Best	Correct?
SPY	Adaptive	Adaptive	Yes
QQQ	Adaptive	Adaptive	Yes
IWM	Binary (t=0)	Binary (t=opt)	No
XLK	Binary (t=opt)	Binary (t=opt)	Yes
XLF	Scaled	Binary (t=0)	No
XLE	Binary (t=0)	Binary (t=0)	Yes

XLP	Binary (t=opt)	Binary (t=opt)	Yes
XLV	Scaled	Scaled	Yes
XLU	Binary (t=opt)	Binary (t=opt)	Yes
GLD	Binary (t=0)	Binary (t=0)	Yes
TLT	Scaled	Scaled	Yes
EEM	Binary (t=0)	Binary (t=0)	Yes
EWJ	Binary (t=0)	Binary (t=0)	Yes
EWZ	Binary (t=0)	Binary (t=0)	Yes
SADAFCO	Binary (t=opt)	Binary (t=opt)	Yes
Al_Rajhi	Scaled	Adaptive	No
STC	Scaled	Scaled	Yes
Mobily	Binary (t=0)	Binary (t=0)	Yes
SABIC	Binary (t=opt)	Binary (t=opt)	Yes
Maaden	Binary (t=opt)	Binary (t=opt)	Yes

Decision tree accuracy: 17/20 = 85%. The 3 misses (IWM, Al Rajhi, XLF) are borderline cases where the second-best strategy is within 0.1 Sharpe of the best.

6.3 Decision Flowchart

Simplified flowchart for strategy selection:

```
START
  |
  v
Is up-week ratio <= 50%?
  YES -> Binary (t=opt): Asset rarely goes up, be selective
  NO  -> Continue
  |
  v
Is price above 50w SMA > 74% of the time?
  YES -> Is regime change frequency <= 5.25/year?
           YES -> Adaptive: Strong stable trend, adapt dynamically
           NO  -> Binary (t=opt): Strong but unstable, be selective
  NO  -> Continue
  |
  v
Is mean-reversion score > 0.07?
  YES -> Scaled: Mean-reverting, size proportionally
  NO  -> Binary (t=0): Random walk, simple directional bet
```

7. When Not to Trade

Not every asset is suitable for this framework. Based on the 20-asset study, the following warning signs indicate an asset should NOT be traded:

Red Flags

- Direction accuracy below 48%: The model fails to predict direction better than random. SADAFCO (48.4%) is at the boundary - it still beats BnH on Sharpe due to drawdown avoidance, but with minimal margin.
- Negative Sharpe excess over BnH for ALL variants: If no strategy improves on passive holding, the model has no predictive edge for this asset.
- Extremely high kurtosis (>50) combined with negative skewness: Catastrophic tail events that the weekly model cannot anticipate (flash crashes, overnight gaps).
- Very low liquidity / sparse trading: Assets with frequent zero-volume weeks or large bid-ask spreads where 0.02% transaction cost is unrealistic.

Asset Suitability Scoring

A simple suitability score can be computed before running the full pipeline:

Score = (Up-Week Ratio - 0.48) x 10
+ (Hurst - 0.45) x 5
+ max(0, Autocorr_Lag1 + 0.3) x 3
- (Kurtosis / 50)

Score > 1.0: Likely suitable for trading
Score 0.5 - 1.0: Marginal, run full pipeline to verify
Score < 0.5: Likely unsuitable, skip or trade with extreme caution

Asset	UpWk	Hurst	AC1	Kurt	Score	Verdict
SPY	0.562	0.521	-0.0814	6.8	1.69	Suitable
QQQ	0.558	0.547	-0.0397	5.1	1.94	Suitable
IWM	0.550	0.582	-0.0508	4.1	2.03	Suitable
XLK	0.556	0.545	-0.0564	2.7	1.91	Suitable
XLF	0.549	0.518	-0.1158	13.5	1.31	Suitable
XLE	0.561	0.513	-0.0115	5.2	1.89	Suitable
XLP	0.563	0.534	-0.0991	5.1	1.75	Suitable
XLV	0.557	0.491	-0.0908	5.8	1.49	Suitable
XLU	0.554	0.516	-0.0623	8.9	1.61	Suitable
GLD	0.560	0.558	-0.0050	2.1	2.18	Suitable
TLT	0.540	0.604	-0.0787	1.5	2.00	Suitable
EEM	0.552	0.590	-0.0529	7.5	2.01	Suitable
EWJ	0.526	0.564	-0.0482	2.9	1.73	Suitable
EWZ	0.535	0.581	-0.0595	3.2	1.86	Suitable
SADAFCO	0.492	0.574	-0.0019	3.1	1.57	Suitable
Al_Rajhi	0.504	0.523	-0.2232	87.5	-0.91	Caution
STC	0.509	0.516	-0.3165	250.9	-4.40	Caution

Mobily	0.504	0.529	0.0237	5.0	1.51	Suitable
SABIC	0.476	0.546	0.0052	4.1	1.27	Suitable
Maaden	0.498	0.487	-0.2644	181.7	-3.16	Caution

8. Practical Application Guide

Step-by-Step for a New Asset

To select the optimal strategy for any new asset:

Step 1: Download at least 3 years of weekly OHLCV data.

Step 2: Compute the 4 key characteristics:
a) Up-week ratio: count(weekly_return > 0) / total_weeks
b) % above SMA-50: count(close > SMA_50) / total_weeks
c) Mean-reversion score: -autocorrelation(returns, lag=1)
d) Regime change freq: count(SMA_20 crossovers) / years

Step 3: Apply the decision tree rules from Section 6.1.

Step 4: Run the full ML pipeline with the selected strategy.

Step 5: Validate with a 52-week rolling backtest. If the selected strategy underperforms BnH, fall back to Binary (t=0) as default.

Fallback Strategy

When in doubt, Binary (t=0) is the safest default. It wins for the most assets (7/20), has the simplest logic (go long if predicted return > 0%, else cash), and never requires threshold optimization. It beats BnH on Sharpe for 18/20 assets.

Portfolio-Level Strategy

For a multi-asset portfolio, consider:

- Assign each asset its optimal strategy per the decision tree
- Use Scaled strategy across the board for maximum drawdown reduction
- Equal-weight allocation across assets for diversification
- Rebalance monthly; re-evaluate strategy assignment annually
- Monitor the suitability score - if it drops below 0.5, stop trading that asset

9. Conclusions

What Determines the Best Strategy?

- Up-week ratio is the single most important discriminator. Assets that go up <50% of weeks need selective entry (Binary t=opt); those with >50% can use simpler strategies.
- Trend strength (% above SMA-50) separates Adaptive from others. Strong trending assets (>74% above SMA-50) with stable regimes benefit from dynamic threshold adjustment.
- Mean-reversion tendency determines Scaled vs Binary. Assets with strong negative autocorrelation should use Scaled (proportional sizing) to avoid overcommitting during reversion periods.
- Regime instability (high regime changes/year) pushes trending assets from Adaptive toward Binary (t=opt) - frequent regime shifts make rolling threshold optimization less stable.

What Does NOT Matter?

- Volatility level alone does NOT predict strategy choice (effect size < 0.3)
- Market (US vs TASI) does NOT predict strategy - the framework is market-agnostic
- Asset class (equity vs bond vs commodity) does NOT determine strategy
- Data length (800 vs 1400 weeks) does NOT meaningfully affect strategy selection

Limitations

- 20 assets is a small sample for decision tree training - rules may not generalize to all markets
- The tree is trained on the same data used for backtesting (no cross-validation of the meta-model)
- Characteristics are static; re-computing on rolling windows could capture regime shifts in the characteristics themselves
- Transaction costs (0.02%) may not apply to all markets and asset sizes
- Currency effects are not hedged for TASI stocks - SAR/USD fluctuation affects USD-denominated portfolios